

WCW Memory Monitor

WinterCrownWORKS · Real-time memory & VRAM tracking plugin for Unreal Engine · v0.2.0 (2026-09-28)

1. Overview

WCW Memory Monitor is a lightweight Slate overlay that lives inside your Unreal project as a **.uplugin**. It is not a standalone app. The HUD reports physical RAM, PC VRAM, the texture streaming pool, per-texture-group resident size, selected LLM tags, and optional RHI resource buckets — plus a build stamp so a capture from a cooked build can be matched to a changelist.

2. Why this plugin

- Catch OOM risk and texture-budget overruns during play, before they reach a packaged player.
- Watch only the texture groups / LLM tags that match your project architecture.
- Toggle the overlay at any time. Heavy RHI walking can be switched off when the frame cost is not worth it.
- Stamp every capture with TitleVersion + CL + UE patch so QA shots are self-describing.

3. UI sample



CitySample. Yellow line is the build stamp from Config/Wcw.ini. Left column is Quality Presets. RHI rows show Use: --- MB (Disabled) after Wcw.MemoryMonitor.EnableRHI 0.

4. Key features

HUD overlay	Semi-transparent Slate widget. Hit-test invisible so gameplay input is not stolen.
System + pool	Physical RAM, DXGI dedicated VRAM (Windows), texture streaming pool required vs. pool size.
Texture groups	Resident mips per TextureGroup, with Project Settings budgets, over-budget tint, and peak hold.
LLM tags	Selected ELLMTag values. Requires launching with -llm.
RHI buckets	Walk of RHIGetTrackedResourceStats (Lumen / Nanite / Shadow / VT / ...). Switch off with EnableRHI when the hitch is too expensive.

Build stamp	Wcw.ini TitleVersion + Changelist, else DefaultGame.ini ProjectVersion, else 1.0.0.0, plus UE patch.
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5. Install: Plugins folder

Copy the repository next to your .uproject. Do not rename the plugin folder.

```

YourProject/
  YourProject.uproject
  Config/
    DefaultEngine.ini
    DefaultGame.ini
    Wcw.ini           ← project stamp (you / CI write this; not inside the plugin)
  Plugins/
    WcwMemoryMonitorModule/
      WcwMemoryMonitorModule.uplugin
      Source/
        Resources/

```

1. Reopen the project. For a C++ project, generate project files and compile.
2. Edit → Plugins → search “WCW Memory Monitor” / “WcwMemoryMonitorModule”, enable, restart if asked.
3. Engine/Plugins/ also works for a shared install. Project Plugins/ is the safer default.

6. Quick start & console commands

Open the in-game console (~) after PIE / Standalone / a packaged Development build.

Wcw.MemoryMonitor.ToggleUI	Show or hide the overlay. Widget is created on first ON.
Wcw.MemoryMonitor.FontScale 1.2	Multiply Project Settings font size. Default 1.0. Applied live on Tick.
Wcw.MemoryMonitor.EnableRHI 0 1	Turn RHI resource tracking off or on. 1 walks tracked RHI resources and fills Lumen / Nanite / Shadow / VT / ... rows. 0 skips the walk and those rows print “Use: --- MB (Disabled)”.

FontScale and most Project Settings persist through DefaultEngine.ini. EnableRHI is a runtime CVar; change it for the session you are looking at.

LLM and runtime environment

LLM: accurate LLM group numbers need the process started with -llm. Without it those rows stay at 0.

PIE: editor play does not match device heaps. Prefer Standalone or a packaged build on target hardware when you sign off a budget.

7. Build stamp — Wcw.ini, fallback, cook / PreBuild

The yellow line is built by MakeBuildLabel() in SWcwMemoryBudgetWidget.cpp. It is resolved once when the widget is constructed (ToggleUI ON).

Resolution order for the version token

1. <Project>/Config/Wcw.ini section [Wcw.Build] key **TitleVersion**
2. Same file, key **ProjectVersion** (alias if TitleVersion is empty)
3. DefaultGame.ini [/Script/EngineSettings.GeneralProjectSettings] **ProjectVersion**
4. Hard fallback 1.0.0.0

If **Changelist** is set and non-empty in Wcw.ini, it is appended as CL Engine patch always comes from FEngineVersion::Current().

Examples

```

Build: 0.0.1 | CL 12345 | UE 5.8.0   ← Wcw.ini present (CitySample capture)
Build: 0.9.3 | CL 88421 | UE 5.5.4   ← CI wrote TitleVersion + Changelist
Build: 1.0.0.0 | UE 5.5.4           ← no Wcw.ini, empty ProjectVersion

```

Wcw.ini template (copy to the GAME project, not the plugin)

```
; WinterCrownWORKS shared project stamp.
; Written by CI / a PreBuild step before cook. The plugin only reads this file.
; Path: <Project>/Config/Wcw.ini

[Wcw.Build]
TitleVersion=0.9.3
Changelist=88421
Stamp=2026-09-28T16:00:00+09:00
Dirty=0

[Wcw.Common]
Studio=WinterCrownWORKS
Project=
```

A file sitting in the plugin repo is invisible at runtime. The reader uses `FPaths::ProjectConfigDir() / "Wcw.ini"`. A missing file fails closed and falls through to Project Settings.

Suggested PreBuild / CI write

Do not hand-edit the changelist on a cook machine. Have the pipeline emit Wcw.ini immediately before UAT / the editor cook. Perforce: P4 changelist. Git: `git rev-list --count HEAD` plus a short SHA, or your title-version script. Unreal can also run a PreBuild step from the `.uproject / Target.cs` for a local editor stamp.

```
# example: GitHub Actions / Jenkins step before RunUAT BuildCookRun
CL=$(git rev-list --count HEAD)
VER=${TITLE_VERSION:-0.9.3}
STAMP=$(date -Iseconds)
cat > "$PROJECT_DIR/Config/Wcw.ini" <<EOF
[Wcw.Build]
TitleVersion=$VER
Changelist=$CL
Stamp=$STAMP
Dirty=0
EOF
# then cook. Confirm Config/Wcw.ini is staged with the packaged game.
```

Packaged builds

A custom ini that is not named Default*.ini is easy to drop from staging if your cook filters Config. Confirm the packaged game still has Config/Wcw.ini. If the file is absent on device, the HUD silently uses ProjectVersion / 1.0.0.0 — that is expected, not a widget bug.

8. Why you may turn RHI tracking off

EnableRHI gates RHIGetTrackedResourceStats. That call walks every tracked RHI resource and string-matches names (Lumen, Nanite, Shadow, DistanceFields, IndexBuffer, VertexBuffer, VirtualTexture, Hair). On a dense scene such as CitySample this is visible in the frame time: the same view that sits near 45 ms with EnableRHI 0 can spike past 100–200 ms for the frames where the walk runs after EnableRHI 1.

- Need numbers: `Wcw.MemoryMonitor.EnableRHI 1` — Lumen / Nanite / Shadow / VT rows fill with MB.
- Need a cheap overlay: `Wcw.MemoryMonitor.EnableRHI 0` — those rows print Use: --- MB (Disabled). Texture / LLM peak-hold still runs.
- Requires a build with `RHI_ENABLE_RESOURCE_INFO`. Test / Development is the intended target; Shipping often has nothing to walk.
- HUD values are non-transient size only.

9. Peak hold and QA video

TextureGroup and LLM rows keep the value from the moment usage crossed budget for about five seconds, as (Peak: 1753.1 MB @ 21:41:06) — **megabytes plus wall-clock time**. Spikes often last a single frame, so a raw Use column looks fine on a screenshot. The hold is what stays in the picture.

When QA later sends only a video, two axes on the same frame line up with the logs:

- The yellow build stamp (TitleVersion / CL / UE) — which drop.
- The Peak timestamp on the red row — when inside that session. It uses `FDateTime::Now()`, the same clock as LogTemp / LLM / memreport lines.

Pause the video, read the Peak hour-minute-second, open the log around that time. One capture then answers which changelist, which second, and which group went over budget. Peak hold on Texture / LLM still runs when RHI tracking is off.

10. Project Settings

Edit → Project Settings → Plugins → WCW Memory Monitor. Changes write to DefaultEngine.ini.

- FontSize — base point size; FontScale CVar multiplies it.
- SystemGroups / TargetGroups / LLMGroups / RhiResourceGroups — which rows appear. Duplicates are remapped in the editor.
- Texture and LLM rows take a BudgetMB used for the red over-budget tint and 5-second peak hold.

Settings walkthrough: <https://www.youtube.com/watch?v=o5P6D3sfGsk>

11. License

MIT License. Use, modify, and redistribute freely. Repository: github.com/yuukajii/UnrealWcwMemoryMonitor
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